Tax Evasion and Productivity*

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Abstract

Value-added taxes (VAT) allow firms to deduct the taxes paid on purchases from the taxes collected on sales, creating an incentive for firms to overreport the purchases of inputs to reduce their tax bill. The fundamental issue in detecting input overreporting stems from two unobservables, true inputs and productivity. Because firms differ in productivity, a firm reporting high input use, for a given level of output, may be either overreporting or simply less productive. To distinguish between overreporting and productivity, I use a structural production-function approach and a benchmark group, a subset of firms that report inputs correctly. Because the benchmark firms are assumed to share the same technology as the other firms, I estimate the technology's parameters from them. Using these estimates, I compare what potentially tax-evading firms report with predictions of the true inputs they would use. I apply the method to a Colombian manufacturing survey from 1981 to 1991. During this period, the Colombian sales tax worked as a VAT and the data include a subset of firms that, due to government and market scrutiny, reported inputs correctly.

My approach yields a simple and robust statistical test to detect input overreporting. Under the null hypothesis of correct reporting, the test is agnostic about the specification of both the probability of detection and the cost of overreporting. Among the 5 largest industries, the test rejects the null hypothesis in all 4 industries subject to the sales tax, and, as expected, it fails to reject in the industry exempt from the sales tax. Firms overreport on average between 4.5 and 23.1 percent of their true inputs, depending on the industry.

The magnitude of detected overreporting matters for methods of estimating production functions that exploit a flexible input to identify productivity (Levinsohn and Petrin 2003; Doraszelski and Jaumandreu 2013; Ackerberg et al. 2015; Gandhi et al. 2020). These methods typically rely on reported materials, the input that firms overreport. I show that failing to correct for overreporting yields larger output elasticities of materials, and more dispersed and less persistent productivity distributions. Using the production-function parameters, I estimate a structural model of input overreporting. With the model, I study how net sales-tax revenue changes as claimed deductions, true and fictitious, respond to the sales-tax rate on purchases. At the current rate, a 1 percent increase in the rate lowers net sales-tax revenue by about 1.7 percent of gross sales-tax revenue. Undetected overreporting costs the government 4.7 percent of the gross sales-tax revenue from potentially tax-evading firms.

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